LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety and emergency devices for retraction and extension, typical u/c circuits
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety and emergency devices for retraction and extension, typical u/c circuits,
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of typical u/c circuit diagrams; identification of safety/emergency devices; instructor supervision of diagram analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear safety valve?
(b) How does the emergency extension system differ from normal retraction?
(c) Name two common safety devices found in typical u/c circuits.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Typical alighting gear construction components • Safety devices for retraction and extension • Emergency extension mechanisms • Typical u/c circuit overview • Hydraulic pressure management in circuits	L & Ds	PPT, IB	15 min
(b)	• Analysis of safety valve operation • Review of emergency extension logic • Identification of circuit breakers • Discussion on hydraulic lockout systems • Interpretation of typical u/c schematics	L & Ds	PPT, PDF	20 min
(c)	• Summary of safety device functions • Review of emergency extension procedures • Key points of typical u/c circuits • Importance of circuit protection devices • Preparation for shock absorber study	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the safety valve in preventing accidental retraction.
(b) Describe the sequence of events during an emergency gear extension.

12. Home Assignment and Follow Up	Prepare a brief note on the shock absorber types and their operation to be discussed in the next lesson on shock observers.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Shock observers, types, brief descriptions, operation of oleos	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min